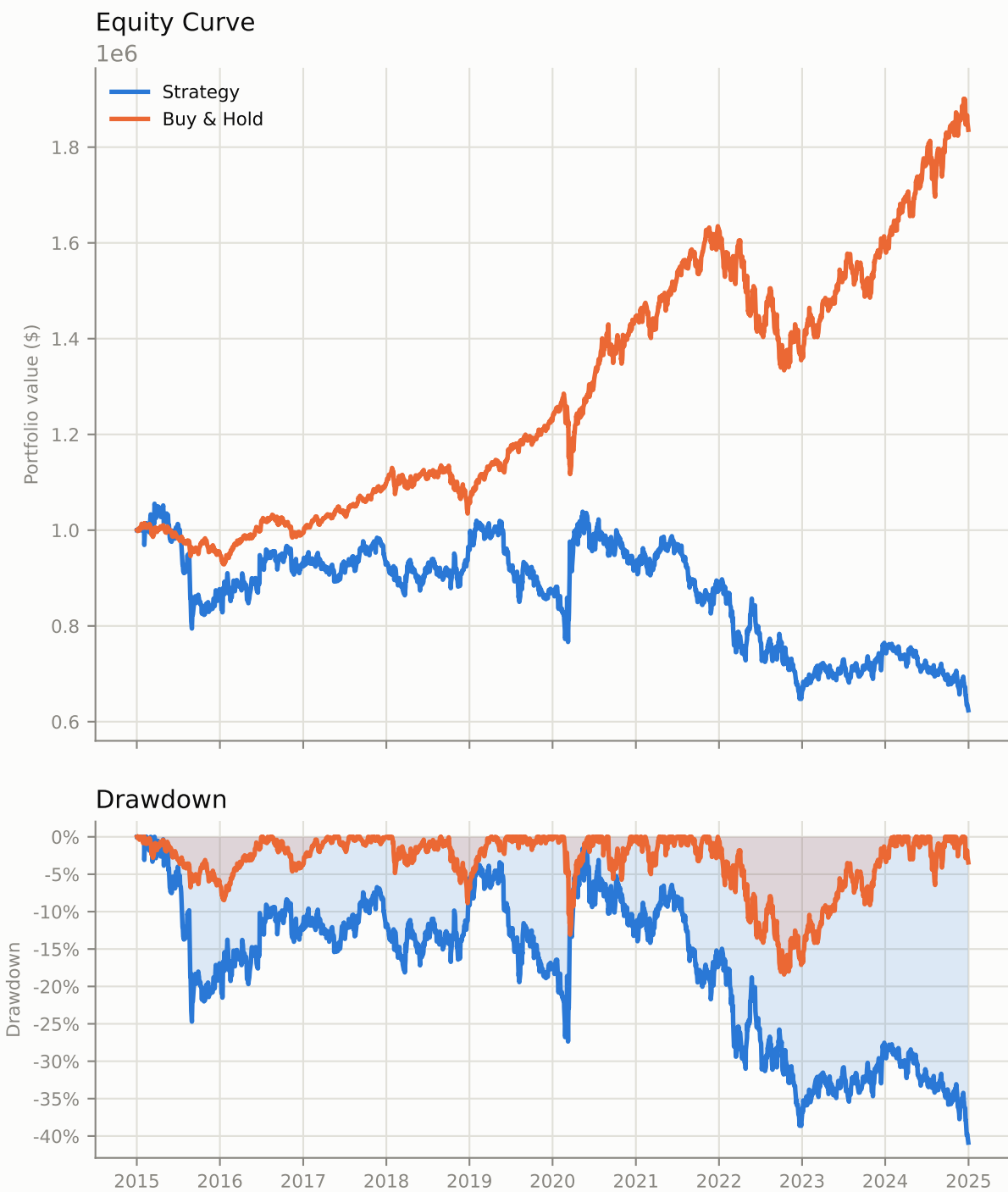


Performance Overview - Score Scaling Neutral



Metric	Strategy	Buy & Hold
Total return	-37.61%	83.62%
Annualized return	-4.61%	6.27%
Annualized vol	14.96%	8.55%
Sharpe	-0.37	0.52
Max drawdown	-40.88%	-18.41%
Drawdown/vol	2.73	2.15
Num trades	14,998	8
Win rate	48.93%	62.50%
Avg win	1,221.93	183,135.61
Avg loss	-1,219.72	-26,577.97
Risk/reward ratio	1.00	6.89
Payoff factor	0.96	11.48
CPC index	0.47	49.46
Time in market	99.17%	99.96%
Annual turnover	688.7x	0.2x

Return Correlation

Strategy	1.00	0.06
Buy & Hold	0.06	1.00
	Strategy	Buy & Hold

Monthly Returns

Strategy

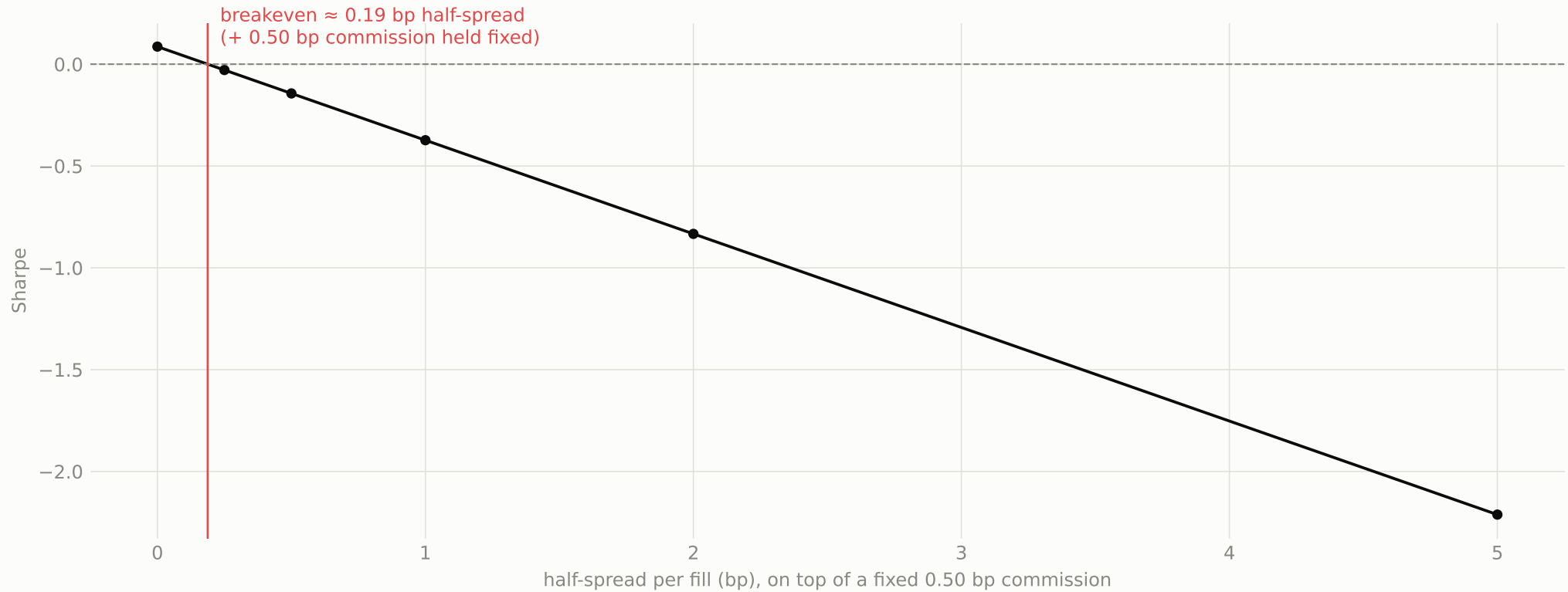
2015	0.0%	0.9%	4.0%	-3.4%	-3.7%	3.6%	-9.5%	-13.3%	8.2%	-2.5%	-0.3%	4.1%	-13.0%	-24.7%	-0.81
2016	2.0%	-1.1%	1.1%	-3.6%	4.6%	2.8%	2.5%	0.5%	0.5%	0.8%	-1.9%	-1.7%	7.7%	-6.9%	0.58
2017	0.7%	-0.1%	-0.9%	-1.1%	-1.8%	3.4%	0.3%	1.0%	1.9%	1.2%	0.8%	-4.9%	-0.9%	-6.4%	-0.05
2018	-1.4%	-2.5%	0.1%	2.3%	-2.8%	2.5%	2.5%	-2.6%	0.4%	1.7%	-1.3%	4.7%	3.2%	-7.3%	0.30
2019	5.3%	-0.4%	0.0%	-0.6%	-3.4%	-4.9%	-1.8%	2.3%	-0.8%	-4.5%	-1.3%	0.8%	-9.3%	-16.6%	-0.69
2020	-2.0%	-9.1%	23.0%	5.2%	2.9%	-5.1%	1.6%	-3.3%	-0.9%	2.1%	-4.5%	-0.7%	6.2%	-13.1%	0.37
2021	2.7%	-2.5%	3.0%	2.8%	-0.9%	-0.4%	-2.3%	-5.8%	-4.1%	1.3%	-1.1%	2.5%	-5.7%	-16.3%	-0.41
2022	-1.8%	-3.4%	-6.3%	-3.0%	9.5%	-6.9%	-3.5%	0.0%	2.1%	-3.8%	-3.6%	-6.7%	-26.4%	-27.8%	-1.58
2023	5.9%	2.0%	1.8%	-4.0%	-0.9%	6.2%	-5.0%	2.7%	-0.2%	1.3%	1.9%	3.8%	13.0%	-6.6%	1.01
2024	-1.2%	0.6%	-2.2%	1.4%	-4.6%	1.2%	-2.1%	0.6%	-2.3%	-1.7%	-0.8%	-8.1%	-17.1%	-18.2%	-1.55
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	Max DD	Sharpe

Buy & Hold

2015	1.1%	0.0%	-1.5%	0.6%	-0.2%	-1.5%	-0.8%	-1.4%	-0.9%	2.5%	-1.6%	-1.3%	-5.0%	-6.7%	-0.91
2016	-0.2%	1.5%	1.8%	0.9%	0.1%	2.4%	1.2%	-0.5%	0.5%	-1.6%	-2.0%	0.4%	4.8%	-4.6%	1.01
2017	1.6%	1.8%	-0.3%	0.8%	1.0%	-0.8%	1.6%	1.5%	-0.6%	1.4%	1.0%	0.9%	10.1%	-2.0%	2.82
2018	2.5%	-1.8%	-0.5%	0.0%	1.7%	-0.4%	0.3%	1.7%	-0.2%	-3.5%	-0.5%	-2.3%	-3.7%	-8.9%	-0.48
2019	3.8%	1.0%	1.6%	1.4%	-2.0%	3.9%	0.6%	1.5%	-0.5%	1.5%	0.8%	1.2%	15.7%	-2.3%	2.90
2020	1.7%	-2.0%	-3.5%	5.5%	2.7%	2.2%	5.0%	3.1%	-2.8%	-2.0%	4.4%	2.8%	16.8%	-13.1%	1.28
2021	-0.8%	-0.5%	0.1%	3.9%	1.0%	1.8%	2.1%	1.5%	-2.9%	4.2%	-0.1%	1.7%	12.8%	-5.0%	1.48
2022	-3.6%	-0.6%	2.1%	-5.9%	-0.4%	-5.0%	4.3%	-3.1%	-6.4%	2.1%	4.1%	-3.9%	-15.8%	-17.9%	-1.16
2023	5.3%	-2.3%	4.6%	0.6%	1.0%	2.7%	2.6%	-1.1%	-3.5%	-0.6%	5.5%	1.7%	17.5%	-5.8%	2.01
2024	0.8%	2.3%	2.5%	-2.1%	3.3%	2.8%	0.4%	1.1%	2.1%	-0.1%	2.3%	-1.7%	15.3%	-6.4%	1.59
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	Max DD	Sharpe

Cost Sensitivity

Sharpe vs half-spread per fill. Commission held fixed at 0.50 bp per fill on every rung, so the whole curve already pays it.



Half-spread (bp)	Total return	Sharpe	Max drawdown
0.00	24.1%	0.09	-27.2%
0.25	4.5%	-0.03	-29.0%
0.50	-12.0%	-0.14	-31.8%
1.00	-37.6%	-0.37	-40.9%
2.00	-68.6%	-0.83	-70.0%
5.00	-96.0%	-2.21	-96.1%

Annual turnover holds at 681-689x across the ladder: this strategy does not trade less when trading costs more, which is why a fraction of a basis point is decisive.

Backtest Configuration

Field	Value
name	Score Scaling Neutral
data	data/raw.parquet
tickers	['SPY', 'QQQ', 'TLT', 'IEF', 'GLD', 'DBC', 'UUP', 'FXE']
portfolio	score_proportional
window	20
initial_cash	1,000,000.0
entry_threshold	0.0
exit_threshold	0.0
vol_window	20
max_gross	2.0
dollar_neutral	True
winsor_limit	3.0
cost_bps	1.0
commission_bps	0.5
drift_band	0.005
stop_loss_pct	None
take_profit_pct	None
max_holding_days	None
output_dir	output
risk_free_rate	0.02